

C Programming — Programs & Problems (Previous Year Papers)

Subject: Problem Solving and Programming in C (B21CA02DC)

Sources: December 2024 | April 2025 | October 2025

Exam section rules (quick reference)

Section	Instruction	Marks each	How many to answer
A	One word or one sentence	1 mark	Any 10
B	Two or three sentences / short program	2 marks	Any 5
C	One paragraph / program / algorithm	4 marks	Any 5
D	Three pages (detailed)	15 marks	Any 2

SECTION A — Output & Short Code (1 Mark Each)

Answer any 10. Program-related questions below.

Q6 — Predict output | 1 mark

Asked in: December 2024

Question: Predict the output of the following program:

```
int main() {  
    int i;  
    for(i=0; i<25; i++);  
    printf("%d", i);  
    return 0;  
}
```

Answer: 25

Why: Semicolon after `for(...)` makes the loop body empty. Loop runs 25 times ($i = 0$ to 24), then i becomes 25. `printf` prints 25.

Q19 — Value of variable | 1 mark

Asked in: April 2025

Question: What will be the value stored in variable `a`?

```
int a = 4.5 + 6.5;
```

Answer: `a = 11`

Why: $4.5 + 6.5 = 11.0$ (double). Assigned to `int a`, the decimal part is truncated → **11**.

Q30 — Predict output | 1 mark

Asked in: April 2025

Question: What will be the output?

```
void main() {  
    register x=10;  
    printf("%d", &x);  
}
```

Answer: **Compilation error**

Why: `register` variables cannot use address operator `&`. Compiler error.

Q35 — Predict output | 1 mark

Asked in: October 2025

Question: Write the output of: `printf("%3d", 0);`

Answer: `0` (0 printed with minimum width 3 — may show spaces before 0)

Why: `%3d` = integer with field width 3. Value 0 fits in 3 characters.

Q21 — for loop syntax | 1 mark ★

Asked in: April 2025 · October 2025

Question: What is the syntax of a for loop in C?

Answer:

```
for(initialization; condition; update)  
    statement;
```

Q22 — Array declaration | 1 mark

Asked in: April 2025

Question: How do you declare an array of 20 integers in C?

Answer: `int arr[20];`

Q36 — do-while loop example | 1 mark

Asked in: December 2024 · October 2025

Question: Give an example for a loop that tests condition at the bottom of the loop.

Answer:

```
do {  
    /* body executes at least once */  
} while (condition);
```

Example: do-while loop — condition checked **after** the body (exit-controlled loop).

SECTION B — Write Programs (2 Marks Each)

Answer any 5. Full program questions below.

Q48 — Conditional operator usage | 2 marks

Asked in: December 2024 · October 2025

Question: Write down the usage of the conditional operator.

Answer:

Syntax: `condition ? value_if_true : value_if_false`

```
int a = 10, b = 20, max;  
max = (a > b) ? a : b; /* assigns larger value */  
printf("Max =%d\n", max);
```

Used as shorthand for simple if-else assignment in one line.

Q56 — Print a message | 2 marks

Asked in: April 2025

Question: Write a C program to print a message.

Answer:

```
#include<stdio.h>
int main() {
    printf("Hello, World!\n");
    return 0;
}
```

Q60 — One dimensional array with example | 2 marks ★

Asked in: December 2024 · April 2025

Question: Define one dimensional array with example.

Answer:

A **one-dimensional array** stores multiple values of the same type in a single row.

```
int marks[5];                /* declaration */
int marks[5] = {10, 20, 30, 40, 50}; /* initialization */
printf("%d", marks[0]);      /* access first element → 10 */
```

Q63 — Swap using call by reference | 2 marks ★🔥

Asked in: December 2024 · April 2025

Question: Write a program to swap two numbers using call by reference.

Answer:

```
#include<stdio.h>
void swap(int *a, int *b) {
    int t = *a;
    *a = *b;
    *b = t;
}
int main() {
    int x, y;
    printf("Enter two numbers: ");
    scanf("%d%d", &x, &y);
    printf("Before swap: x=%d, y=%d\n", x, y);
    swap(&x, &y);
    printf("After swap: x=%d, y=%d\n", x, y);
    return 0;
}
```

Q65 — Significance of argc and argv | 2 marks ★🔥

Asked in: December 2024 · April 2025 · October 2025

Question: What is the significance of argc and argv in command line arguments?

Answer:

- **argc** — argument count (includes program name)
- **argv** — array of strings; `argv[0]` = program name, `argv[1]`, `argv[2]` = user inputs
- Lets program read inputs at run time without recompiling

```
int main(int argc, char *argv[]) {
    printf("Total args =%d\n", argc);
    printf("Program name =%s\n", argv[0]);
    return 0;
}
/* Run: ./prog hello → argc=2, argv[0]=./prog, argv[1]=hello */
```

Q68 — if-else-if ladder | 2 marks ★🔥

Asked in: April 2025 · October 2025

Question: Explain if—else if ladder.

Answer:

Tests multiple conditions **top to bottom**. First true condition runs; rest are skipped. Optional final `else` handles no match.

```
int marks;
printf("Enter marks: ");
scanf("%d", &marks);
if(marks >= 90)
    printf("Grade A\n");
else if(marks >= 75)
    printf("Grade B\n");
else if(marks >= 60)
    printf("Grade C\n");
else
    printf("Fail\n");
```

Q69 — Print 1 to 10 | 2 marks

Asked in: October 2025

Question: Write a program to print 1 to 10 using for loop.

Answer:

```
#include<stdio.h>
int main() {
    int i;
    for(i = 1; i <= 10; i++)
        printf("%d ", i);
    return 0;
}
```

Q70 — Odd or even using function | 2 marks

Asked in: October 2025

Question: Write a program to check whether a number is odd or even using user defined function.

Answer:

```
#include<stdio.h>
int check(int n) {
    if(n % 2 == 0)
        return 1; /* even */
    else
        return 0; /* odd */
}
int main() {
    int n;
    printf("Enter a number: ");
    scanf("%d", &n);
    if(check(n))
        printf("%d is Even\n", n);
    else
        printf("%d is Odd\n", n);
    return 0;
}
```

Q72 — Predict output | 2 marks

Asked in: October 2025

Question: Write the output of the following code:

```
void main() {
    printf("Hai");
    main();
    return 0;
}
```

Answer: Prints `Hai` repeatedly many times, then **stack overflow / program crash**.

Why: `main()` calls itself — infinite recursion with no base case.

SECTION C — Programs, Algorithms, Flowcharts (4 Marks Each)

Answer any 5. Program / algorithm / flowchart questions below.

Q78 — Entry-controlled vs exit-controlled loops | 4 marks



Asked in: December 2024 · October 2025

Question: Explain entry-controlled loop and exit-controlled loop.

Answer:

Type	When condition checked	Loops	Min runs
Entry-controlled	Before body	for, while	0 times
Exit-controlled	After body	do-while	1 time

Entry-controlled (while):

```
while(i <= 10) {
    printf("%d ", i);
    i++;
}
```

Exit-controlled (do-while):

```
do {
    printf("%d ", i);
    i++;
} while(i <= 10);
```

Q80 — Call by value vs reference with example | 4 marks



Asked in: December 2024 · April 2025 · October 2025

Question: Differentiate call by value and call by reference with example.

Answer (theory + swap example):

Call by value	Call by reference
Copy of value passed	Address passed using pointer
Original unchanged	Original changed
<code>func(a)</code> with <code>int x</code>	<code>func(&a)</code> with <code>int *x</code>

```
void swap(int *a, int *b) {  
    int t = *a; *a = *b; *b = t;  
}  
/* main: swap(&x, &y); */
```

Q86 — Algorithm sum 1 to 100 | 4 marks

Asked in: April 2025

Question: Write an algorithm to find the sum of the first 100 natural numbers.

Answer:

```
1. START  
2. SET sum = 0, i = 1  
3. WHILE i <= 100 DO  
    sum = sum + i  
    i = i + 1  
END WHILE  
4. PRINT sum  
5. STOP
```

Result: sum = 5050

Q89 — break vs continue | 4 marks



Asked in: April 2025 · October 2025

Question: Write down the difference between break and continue statement.

Answer:

break	continue
Exits loop/switch completely	Skips rest of current iteration only
Control goes after loop	Goes to next iteration
Used in switch to avoid fall-through	Used inside loops only

```
for(i = 1; i <= 5; i++) {
    if(i == 3) continue; /* skip printing 3 */
    if(i == 5) break;    /* stop at 5 */
    printf("%d ", i);
}
/* Output: 1 2 4 */
```

Q90 — Dynamic memory functions with examples | 4 marks

Asked in: December 2024 · April 2025

Question: Explain the functions malloc(), calloc(), realloc(), and free() in C with examples.

Answer:

Function	Purpose
malloc(size)	Allocates memory; not initialized
calloc(n, size)	Allocates n blocks; initialized to 0
realloc(ptr, size)	Resizes allocated memory
free(ptr)	Releases memory

```
#include<stdlib.h>
int *arr = (int*)malloc(5 * sizeof(int));
if(arr == NULL) { printf("Failed"); exit(1); }
arr[0] = 10;
free(arr); /* always free after use */
```

Q91 — Sum of squares using do-while | 4 marks

Asked in: April 2025

Question: Write a program to find sum of squares of n natural numbers using do while loop.

Answer:

```
#include<stdio.h>
int main() {
    int n, i = 1, sum = 0;
    printf("Enter n: ");
    scanf("%d", &n);
    do {
        sum = sum + (i * i);
        i++;
    } while(i <= n);
    printf("Sum of squares =%d\n", sum);
    return 0;
}
```

Q93 — Compare Structure and Union with example | 4 marks

Asked in: April 2025

Question: Compare Structure and Union with example.

Answer:

Feature	Structure	Union
Memory	Separate for each member	Shared by all members
Size	Sum of members	Largest member only
Members active	All at same time	One at a time

```
struct student {
    char name[20];
    int roll;
    float marks;
} s1;

union data {
    int i;
    float f;
} d;
d.i = 10;
d.f = 3.14;    /* overwrites i */
```

Q95 — Flowchart biggest of two | 4 marks

Asked in: April 2025

Question: Draw a flow chart to find biggest among two numbers.

Answer (algorithm steps):

```
START → Input A, B → Is A > B?  
YES → Print A → STOP  
NO → Print B → STOP
```

Symbols: Oval=Start/Stop | Parallelogram=Input | Diamond=Decision |
Rectangle=Process

Q96 — Type conversion with example | 4 marks

Asked in: December 2024 · October 2025

Question: Explain type conversion with example.

Answer:

Implicit conversion — done automatically by compiler:

```
int a = 3.7;          /* float → int, a = 3 */  
float b = 5;          /* int → float, b = 5.0 */
```

Explicit conversion (casting):

```
float result = (float)9 / 2; /* result = 4.5 */  
int x = 9 / 2;               /* x = 4 (integer division) */
```

Converting larger type to smaller may lose data (truncation).

Q97 — Largest of 3 using conditional operator | 4 marks



Asked in: December 2024 · October 2025

Question: Write a program to find the largest of three using conditional operator.

Answer:

```
#include<stdio.h>  
int main() {  
    int a, b, c, max;  
    printf("Enter three numbers: ");  
    scanf("%d%d%d", &a, &b, &c);
```

```

max = (a > b) ? ((a > c) ? a : c) : ((b > c) ? b : c);
printf("Largest =%d\n", max);
return 0;
}

```

Q99 — Positive difference | 4 marks

Asked in: October 2025

Question: Write a program to print the positive difference between two numbers.

Answer:

```

#include<stdio.h>
int main() {
    int a, b, diff;
    printf("Enter two numbers: ");
    scanf("%d%d", &a, &b);
    if(a > b)
        diff = a - b;
    else
        diff = b - a;
    printf("Positive difference =%d\n", diff);
    return 0;
}

```

Q100 — Switch case with example | 4 marks ★🔥

Asked in: April 2025 · October 2025

Question: Explain switch case with an example.

Answer:

```

#include<stdio.h>
int main() {
    int choice;
    printf("Enter 1-Add, 2-Sub, 3-Mul: ");
    scanf("%d", &choice);
    switch(choice) {
        case 1: printf("Add selected\n"); break;
        case 2: printf("Sub selected\n"); break;
        case 3: printf("Mul selected\n"); break;
        default: printf("Invalid\n");
    }
}

```

```
}  
    return 0;  
}
```

Q101 — break and continue with examples | 4 marks

Asked in: October 2025

Question: Explain the difference between break and continue with examples.

Answer:

break — exits the loop immediately:

```
for(i = 1; i <= 10; i++) {  
    if(i == 6) break;  
    printf("%d ", i);  
}  
/* Output: 1 2 3 4 5 */
```

continue — skips current iteration, continues loop:

```
for(i = 1; i <= 5; i++) {  
    if(i == 3) continue;  
    printf("%d ", i);  
}  
/* Output: 1 2 4 5 */
```

SECTION D — Long Programs & Detailed Notes (15 Marks Each)

Answer any 2. Program questions below.

Q106 — Algorithm, flowchart and symbols | 15 marks 🔥

Asked in: December 2024

Question: Write a detailed note on the algorithm, flowchart, and symbols used in the flowchart with example.

Answer:

Algorithm: Step-by-step finite procedure to solve a problem.

Example — find largest of two numbers:

1. START
2. Input A, B
3. If $A > B$ then Print A else Print B
4. STOP

Flowchart symbols:

Symbol	Name	Use
Oval	Terminal	Start / Stop
Rectangle	Process	Calculation, assignment
Parallelogram	Input/Output	Read / Print
Diamond	Decision	if / condition
Arrow	Flow line	Direction of flow

Flow: START → Input A,B → Is $A > B$? → YES: Print A → STOP | NO: Print B → STOP

Q107 — Loop control structures with examples | 15 marks



Asked in: December 2024 · October 2025

Question: Explain different loop control structures used in C programs with examples.

Answer: Explain **for**, **while**, **do-while** with syntax and one example each.

for loop:

```
for(i = 1; i <= 5; i++)  
    printf("%d ", i);
```

while loop:

```
i = 1;  
while(i <= 5) {  
    printf("%d ", i);  
    i++;  
}
```

do-while loop:

```
i = 1;  
do {  
    printf("%d ", i);
```

```
i++;
} while(i <= 5);
```

Loop	Type	Runs at least once?
for	Entry-controlled	No
while	Entry-controlled	No
do-while	Exit-controlled	Yes

Q108 — Recursion with example program | 15 marks ★🔥

Asked in: December 2024 · April 2025 · October 2025

Question: Explain recursion and its types with an example program.

Answer:

Definition: A function that calls itself. Must have **base case** (stop) and **recursive case**.

Types: Direct recursion (calls itself) | Indirect recursion (A calls B, B calls A)

Example — Factorial:

```
#include<stdio.h>
int fact(int n) {
    if(n <= 1)          /* base case */
        return 1;
    return n * fact(n - 1); /* recursive case */
}
int main() {
    int n = 5;
    printf("Factorial of %d = %d\n", n, fact(n));
    return 0;
}
```

Output: Factorial of 5 = 120

Q109 — Sort array ascending | 15 marks

Asked in: December 2024

Question: Write a C program to read n array elements and sort in ascending order.

Answer:

```
#include<stdio.h>
int main() {
```

```

int n, a[100], i, j, temp;
printf("Enter n: ");
scanf("%d", &n);
printf("Enter%d elements: ", n);
for(i = 0; i < n; i++)
    scanf("%d", &a[i]);
/* Bubble sort */
for(i = 0; i < n-1; i++)
    for(j = 0; j < n-i-1; j++)
        if(a[j] > a[j+1]) {
            temp = a[j];
            a[j] = a[j+1];
            a[j+1] = temp;
        }
printf("Sorted array: ");
for(i = 0; i < n; i++)
    printf("%d ", a[i]);
return 0;
}

```

Q110 — Operators in C with examples | 15 marks ★🔥

Asked in: April 2025 · October 2025

Question: Explain various operators used in C with example.

Answer:

Type	Operators	Example
Arithmetic	+, -, *, /, %	9 % 2 = 1
Relational	==, !=, <, >, <=, >=	a > b
Logical	&&, , !	(a>0 && b>0)
Assignment	=, +=, -=	a += 5
Increment	++, --	i++
Conditional	? :	max = (a>b)?a:b
Bitwise	&, , ^, ~	a & b

```

int a = 10, b = 3;
printf("Sum=%d Mod=%d\n", a+b, a%b);
printf("Max=%d\n", (a>b)?a:b);

```

Q111 — if-else-if ladder and switch with example | 15 marks

Asked in: April 2025 · October 2025

Question: Discuss about if-else-if ladder and switch statement with example.

Answer:

if-else-if ladder — multiple conditions tested in order:

```
if(marks >= 90)      printf("A");
else if(marks >= 75) printf("B");
else if(marks >= 60) printf("C");
else                 printf("Fail");
```

switch — multi-way branch on single value:

```
switch(choice) {
    case 1: printf("Add"); break;
    case 2: printf("Sub"); break;
    case 3: printf("Mul"); break;
    default: printf("Invalid");
}
```

Use if-else-if for ranges; switch for fixed values. Always use `break` in switch.

Q112 — Average of 10 numbers using array | 15 marks

Asked in: December 2024 · April 2025

Question: Discuss different types of arrays with examples, and write a program to find the average of 10 numbers using a one-dimensional array.

Answer:

1D array: `int arr[10];` — single row

2D array: `int mat[3][4];` — rows and columns

Program — Average of 10 numbers:

```
#include<stdio.h>
int main() {
    int arr[10], i;
    float sum = 0, avg;
    printf("Enter 10 numbers:\n");
```

```

for(i = 0; i < 10; i++) {
    scanf("%d", &arr[i]);
    sum = sum + arr[i];
}
avg = sum / 10;
printf("Average =%.2f\n", avg);
return 0;
}

```

Q113 — Storage classes with examples | 15 marks ★🔥

Asked in: December 2024 · April 2025 · October 2025

Question: Describe storage classes with examples.

Answer:

Class	Scope	Lifetime	Default value
auto	Block	Block	Garbage
register	Block	Block	Garbage
static	Block/File	Program	0
extern	Program	Program	0

auto — default for local variables:

```

void func() {
    int x;    /* auto by default */
}

```

static — retains value between calls:

```

void counter() {
    static int count = 0;
    count++;
    printf("%d ", count);    /* prints 1, 2, 3 on each call */
}

```

extern — uses global variable defined elsewhere.

Q116 — Largest & second largest using function | 15 marks

Asked in: October 2025

Question: Write a program to find the largest and second largest element in an array using function.

Answer:

```
#include<stdio.h>
void findMax(int arr[], int n, int *max, int *second) {
    int i;
    *max = *second = arr[0];
    for(i = 1; i < n; i++) {
        if(arr[i] > *max) {
            *second = *max;
            *max = arr[i];
        } else if(arr[i] > *second && arr[i] != *max)
            *second = arr[i];
    }
}
int main() {
    int a[100], n, i, max, second;
    printf("Enter n: ");
    scanf("%d", &n);
    for(i = 0; i < n; i++)
        scanf("%d", &a[i]);
    findMax(a, n, &max, &second);
    printf("Largest =%d, Second largest =%d\n", max, second);
    return 0;
}
```

Q117 — File copy using command line arguments | 15 marks

Asked in: April 2025 · October 2025

Question: Write a program to implement file copy using command line arguments.

Answer:

```
#include<stdio.h>
#include<stdlib.h>
int main(int argc, char *argv[]) {
    FILE *src, *dest;
    char ch;
    if(argc != 3) {
        printf("Usage:%s source dest\n", argv[0]);
    }
}
```

```

        return 1;
    }
    src = fopen(argv[1], "r");
    dest = fopen(argv[2], "w");
    if(src == NULL || dest == NULL) {
        printf("Error opening file\n");
        return 1;
    }
    while((ch = fgetc(src)) != EOF)
        fputc(ch, dest);
    fclose(src);
    fclose(dest);
    printf("File copied successfully\n");
    return 0;
}

```

Run: `./copy source.txt dest.txt`

MASTER TABLE — All Program-Related PYQs

Q#	Marks	Year(s)	Type	Priority
Q6	1	Dec 2024	Output prediction	🔥
Q19	1	Apr 2025	Output / type conversion	
Q21	1	Apr 2025, Oct 2025	for loop syntax	★
Q22	1	Apr 2025	Array declaration	
Q30	1	Apr 2025	Output / register error	🔥
Q35	1	Oct 2025	Output / printf format	
Q36	1	Dec 2024, Oct 2025	do-while loop example	★🔥
Q48	2	Dec 2024, Oct 2025	Conditional operator usage	★
Q56	2	Apr 2025	Write program — hello	
Q60	2	Dec 2024, Apr 2025	1D array with example	★
Q63	2	Dec 2024, Apr 2025	Write program — swap	★🔥
Q65	2	Dec 2024, Apr 2025, Oct 2025	argc and argv significance	★🔥

Q#	Marks	Year(s)	Type	Priority
Q68	2	Apr 2025, Oct 2025	if-else-if ladder	★🔥
Q69	2	Oct 2025	Write program — 1 to 10	
Q70	2	Oct 2025	Write program — odd/even	
Q72	2	Oct 2025	Output — recursion	🔥
Q78	4	Dec 2024, Oct 2025	Entry vs exit controlled loops	★🔥
Q80	4	Dec 2024, Apr 2025, Oct 2025	Call by value/ref + example	★🔥
Q86	4	Apr 2025	Algorithm — sum 1–100	
Q89	4	Apr 2025, Oct 2025	break vs continue	★🔥
Q90	4	Dec 2024, Apr 2025	malloc/calloc/realloc/free	★🔥
Q91	4	Apr 2025	Program — sum of squares	
Q93	4	Apr 2025	Structure vs Union + example	
Q95	4	Apr 2025	Flowchart — biggest of two	
Q96	4	Dec 2024, Oct 2025	Type conversion + example	★🔥
Q97	4	Dec 2024, Oct 2025	Program — largest of 3	★
Q99	4	Oct 2025	Program — positive difference	
Q100	4	Apr 2025, Oct 2025	Switch case + example	★🔥
Q101	4	Oct 2025	break and continue + examples	
Q106	15	Dec 2024	Algorithm, flowchart, symbols	🔥
Q107	15	Dec 2024, Oct 2025	Loops with examples	★🔥
Q108	15	Dec 2024, Apr 2025, Oct 2025	Recursion + program	★🔥
Q109	15	Dec 2024	Sort array ascending	

Q#	Marks	Year(s)	Type	Priority
Q110	15	Apr 2025, Oct 2025	Operators with examples	★🔥
Q111	15	Apr 2025, Oct 2025	if-else-if + switch + example	★🔥
Q112	15	Dec 2024, Apr 2025	Average of 10 + arrays	★🔥
Q113	15	Dec 2024, Apr 2025, Oct 2025	Storage classes + examples	★🔥
Q116	15	Oct 2025	Largest & 2nd largest	
Q117	15	Apr 2025, Oct 2025	File copy argc/argv	★🔥

Wishing you all success in your semester exams.

Dua mein Yaad Rakhna.